LAB # 02

INTRODUCTION TO TOP-DOWN APPROACH AND DATA FLOW LEVEL MODELING

Spring 2024

CSE-308L Digital System Design Lab

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"On my honor, as student at University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work."

Submitted to:

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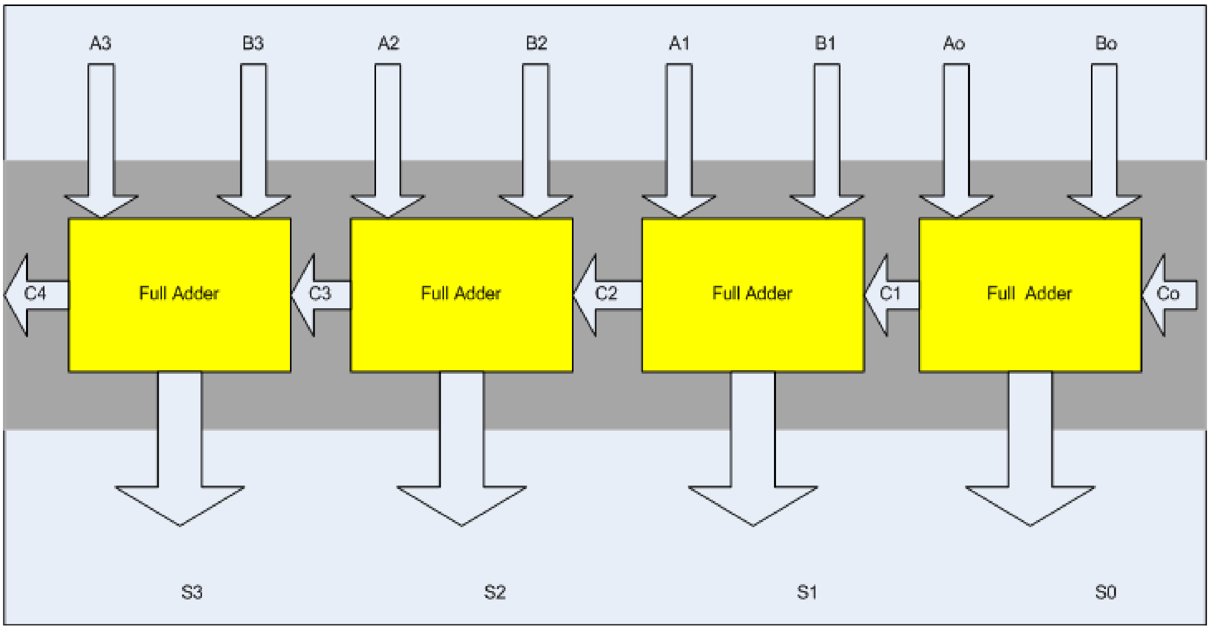
Objectives of The Lab:

- Learn top down and bottom-up design methodologies
- Data flow level modeling

Lab Tasks:

Task 1a:

<u>4-bit Ripple Carry Adder</u>



1- First implement a Full adder using data gate level modeling.
2- Simulate the Full adder with a test bench.
3- Instantiate the Full adder four times and connect the circuit as shown.
4- Now again write a test bench and simulate the 4 bit RCA.

1. Full Adder:

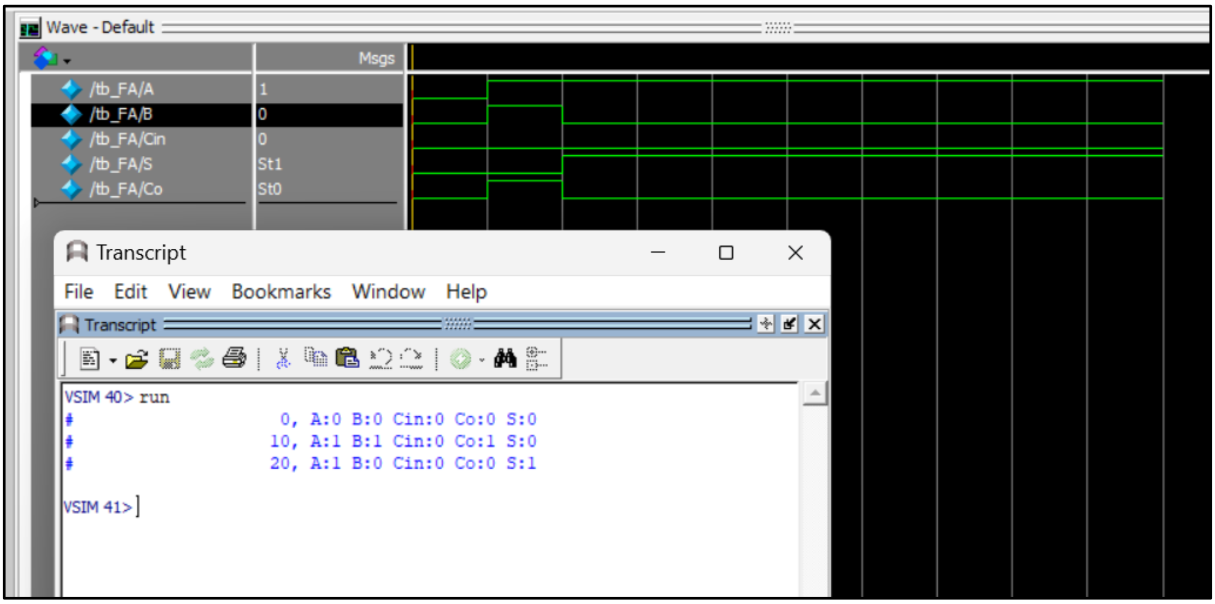
DUT:

```
module FA(S,Co,A,B,Cin);
    input A,B,Cin;
    output S,Co;
    wire y1,y2,y3,y4;
    //S = Cin XOR (A XOR B)
    xor x1(y1,A,B);
    xor x2(S,y1,Cin);

    //Co = AB + Cin.(A XOR B)
    xor x3(y2,A,B);
    and a1(y3,y2,Cin);
    and a2(y4,A,B);
    or o1(Co,y3,y4);
endmodule
```

2. FA Test Bench:

```
module tb_FA();
    reg A,B,Cin;
    wire S,Co;

FA f1(S,Co,A,B,Cin);
initial begin
         A=0; B=0; Cin=0;
    #10; A=1; B=1; Cin=0;
    #10; A=1; B=0; Cin=0;
end
    initial
        $monitor("%d,  A:%b B:%b Cin:%b Co:%b S:%b",$time,A,B,Cin,Co,S);
endmodule
```

Output:



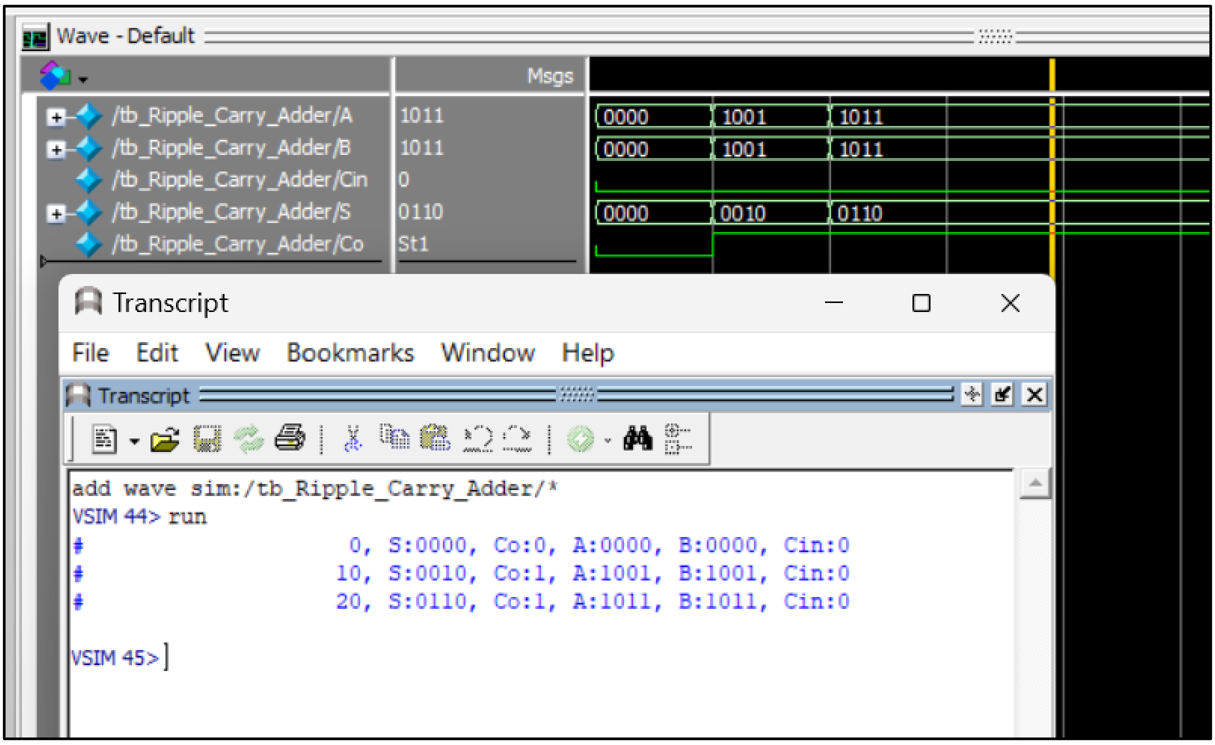
3. RCA DUT

```
module Ripple_Carry_Adder(S,Co,A,B,Cin);
    input [3:0]A,B;
    input Cin;
    output [3:0]S;
    output Co;
    wire c1,c2,c3;
    FA fa1(S[0],c1,A[0],B[0],Cin);
    FA fa2(S[1],c2,A[1],B[1],c1);
    FA fa3(S[2],c3,A[2],B[2],c2);
    FA fa4(S[3],Co,A[3],B[3],c3);
endmodule
```

4. 4-bit RCA Test Bench

```
module tb_Ripple_Carry_Adder();
reg [3:0]A,B;
reg Cin;
wire [3:0]S;
wire Co;

Ripple_Carry_Adder RCA1(S,Co,A,B,Cin);

initial begin
          A=4'b0000; B=4'b0000; Cin=0;
    #10; A=4'b1001; B=4'b1001; Cin=0;
    #10; A=4'b1011; B=4'b1011; Cin=0;


end
    //initial
    //  $monitor("S, Co, A, B, Cin",S,Co,A,B,Cin);
    initial
        $monitor("%d, S:%b, Co:%b, A:%b, B:%b, Cin:%b",$time,S,Co,A,B,Cin);
endmodule
```

Output:



Task 1b:

Design the 4 bit full adder using data flow level modeling.

DUT:

```
module Full_Adder_Data_Flow(S,Co,A,B,Cin);
input A,B,Cin;
output S,Co;

assign S = A^B^Cin;
assign Co = A&B | Cin&(A^B);

endmodule
```